Problem Statement 1: Store Performance and Workforce Productivity – Framing Note

Business Analyst, Global Retail Analytics | For: Head of Store Operations | Tool: Excel (Power Query, Power Pivot)

Evidence base: 35 stores in 7 countries, 1,746 employees and 3,340,935 transaction lines, 1 Jan 2023 – 18 Mar 2025. All amounts in USD (1 EUR = 1.08, 1 GBP = 1.27, 1 CNY = 0.14); returns are netted out of sales. Headcount comes from the employees table, because the stated Number of Employees disagrees with it for 5 stores. Figures are a first look at the data; Part B tests them.

Situation. Zara's 35 stores made $57.5M in net sales in 2023 and $58.2M in 2024. **Complication.** Sales per employee range from $121,111 (Fifth Avenue) to $40,137 (Zeil), a 3× gap. Spain, the home market, grew 2.3% in 2024 while the US grew 14.9% and China fell 12.4%. Footfall is uneven: Saturday carries 22–29% of a week's sales, and the German stores close on Sundays. **Question.** Which stores underperform and why, and are staff in the right places at the right times?

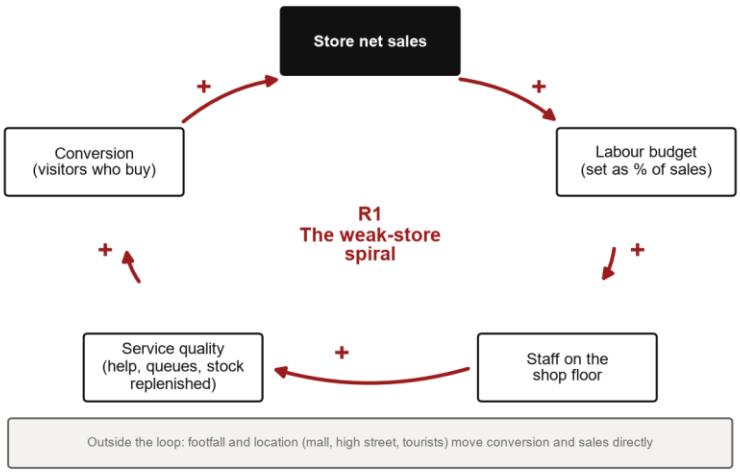
Problem statement. The Head of Store Operations is facing a 3× gap in sales per employee between stores while trying to decide where to put staff hours, which results in weak stores being cut back and falling further behind. **How might we** help store managers put their people on the busiest days, so that weaker stores sell more per employee without adding labour cost?

1. Issue tree

"Why do some stores sell far less than others?" splits into five branches that do not overlap (MECE). Each ends in something Excel can measure.

Branch	Sub-question	Measure in Part B	First look
Market and location	Does the country or city explain the gap?	Net sales by country; each store vs its country average	US +14.9%, China −12.4% (2024)
Number of transactions	Do weak stores serve fewer customers per employee?	Invoices per employee	1,857 at Fifth Avenue vs 495 at Zeil
Basket size	Do customers in weak stores spend less per visit?	Average basket value; units per invoice	$65 at Fifth Avenue vs $81 at Zeil
Staffing	Does headcount follow sales? Where are staff short or surplus?	Sales per employee; staff gap vs company average	Correlation 0.89; gaps from +19 to −34 staff
Product mix	Do weak stores sell a different mix, or lose more to returns?	Category mix; return rate by store	Germany returns 15.5% of sales vs 10.9% overall

2. Systems thinking: the feedback loop and the iceberg



Causal loop. + = the two move the same way; − = opposite ways; R = reinforcing loop.

- **Reinforcing loop (R), a vicious cycle.** Lower sales → smaller labour budget → fewer staff on the floor → slower service and longer queues → fewer browsers become buyers → lower sales. Left alone, a weak store keeps sinking.
- **Leverage point.** The roster, not the headcount. Putting staff on Friday and Saturday, when 36–48% of a week's sales happen, can lift conversion without adding cost. Location and footfall sit outside the loop.

Iceberg level	In this problem
Event (what we see)	Zeil sells $40K per employee, a third of Fifth Avenue
Pattern (to confirm)	The same stores stay at the bottom every month; the gap is widest on Fridays and Saturdays
Structure	Labour budgets follow last period's sales; rosters spread hours evenly across the week
Mental model	"A quiet store needs fewer staff", which may cut staff exactly when customers come in

3. Stakeholder map (power–interest grid)

	Lower interest in store productivity	Higher interest in store productivity
Higher power over staffing	**Keep satisfied: HR.** Needs evidence before any hiring or roster change.	**Manage closely: regional and store managers.** Need a fair ranking and guidance they can act on in their own store.
Lower power over staffing	**Monitor: customers.** Feel the result as queues and service.	**Keep informed: sales associates.** Need targets that do not punish a quiet shift.

4. User research

Method: semi-structured interviews with store managers (generative research). Questions ask about past weeks, not hypothetical ones, and avoid leading wording. No interviews have been held yet, so both personas are drafts to validate.

	Store manager (draft)	Frequent shopper (draft)
Who	Runs one store with 24–106 staff; reports to a regional manager	Shops in store every few weeks, mostly for new arrivals
Goal	Hit the sales target inside the labour budget	Find the right size quickly and leave
Pain points	Weekly numbers arrive late and mix sales with returns	Queues and nobody free in the fitting rooms on Saturdays
Job to be done	When the week looks off, I want to see which days I was short, so I can fix next week's roster	When I have little time, I want help fast, so I can buy and go

#	Interview question (store managers)	What it tests
1	Walk me through last Saturday. When did it get busy, and how many people were on the floor?	Peak-day staffing
2	How did you decide who worked which shift last week? What did you look at?	How rosters are made
3	Think of your best week this year. What was different from a bad week?	Drivers the data cannot see
4	The last time you were short-staffed, what did customers stop getting first?	The service → conversion link
5	Which numbers do you check each week, and which do you not trust?	Scorecard requirements
6	If you could change one rule about staffing or targets, what would it be?	Constraints and quick wins

5. Metric tree

North Star: net sales per employee per month (USD). It joins what a store earns with what it spends on people. It is a lagging indicator, so the tree breaks it into leading inputs a store can move this week.

Level	Metric	Calculation	Type	2024 baseline
North Star	Net sales per employee per month	Net sales ÷ employees ÷ months	Lagging	$2,779
Input	Invoices per employee	Purchase invoices ÷ employees	Leading	33 a month
Input	Average basket value	Net sales ÷ purchase invoices	Leading	$84.57
Input	Units per invoice	Net units ÷ purchase invoices	Leading	2.0
Quality check	Return rate	Returned value ÷ gross sales	Leading	11.2%

Net sales per employee = invoices per employee × average basket. Return rate sits beside it so a store cannot look good on sales that come back.

6. Hypotheses for Part B

#	We believe that… because…	We will know it is true when…	Test in Excel
H1	Most of the gap comes from the number of invoices, not the basket, because flagships get more passers-by	Invoices per employee vary more across stores than basket value	Invoices per employee vs basket by store
H2	Stores with more staff per invoice do not get a bigger basket, because extra staff are not on the floor at peak	Basket does not rise with staff per invoice	Staff per invoice vs basket
H3	Weak stores lose most on Fridays and Saturdays, because that is when staff are stretched	Their gap to strong stores is widest on those days	Sales by store and weekday
H4	A few associates make most of a store's sales	The best associate sells far more than the weakest	Spread of sales per associate
H5	Return rate is higher in the weakest stores	Return rate falls as sales per	Return rate vs sales per

		employee rises	employee
H6	German stores look weaker partly because they trade six days	The gap shrinks per trading day	Sales per trading day

Assumptions and limits

- No opening hours, footfall, floor size or wages in the data, so "overstaffed" means more staff than sales justify.
- Store 12 (Mönckebergstraße, Hamburg) has no sales in Feb–Mar 2024; 1,452 lines were served by staff from another store and stay with the selling store.
- The data is simulated for teaching and does not represent Zara's actual stores or results.